

$$\coth n = \frac{1}{\tanh n} \rightarrow 1$$

$$\tanh n = \frac{\sinh n}{\cosh n}$$

$$= \frac{e^n - e^{-n}}{e^n + e^{-n}}$$

$$= \frac{1 - e^{-2n}}{1 + e^{-2n}} \rightarrow 1$$

$$n^{\frac{1}{n}} = \sqrt[n]{n} \rightarrow 1$$

$$\sinh n = \frac{e^n - e^{-n}}{2}$$

$$\cosh n = \frac{e^n + e^{-n}}{2}$$

$$\operatorname{sech} n = \frac{1}{\cosh n}$$

$$= \frac{2}{e^n + e^{-n}}$$

$$\rightarrow 0$$

Ex 9.4 Q51

$$\sqrt[n]{n} \rightarrow 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n \sqrt[n]{n}} \approx \frac{1}{n}$$

ignore it

(LCT) $a_n = \frac{1}{n \sqrt[n]{n}}$, $b_n = \frac{1}{n}$

$$\lim \frac{a_n}{b_n} = \dots = 1 > 0$$

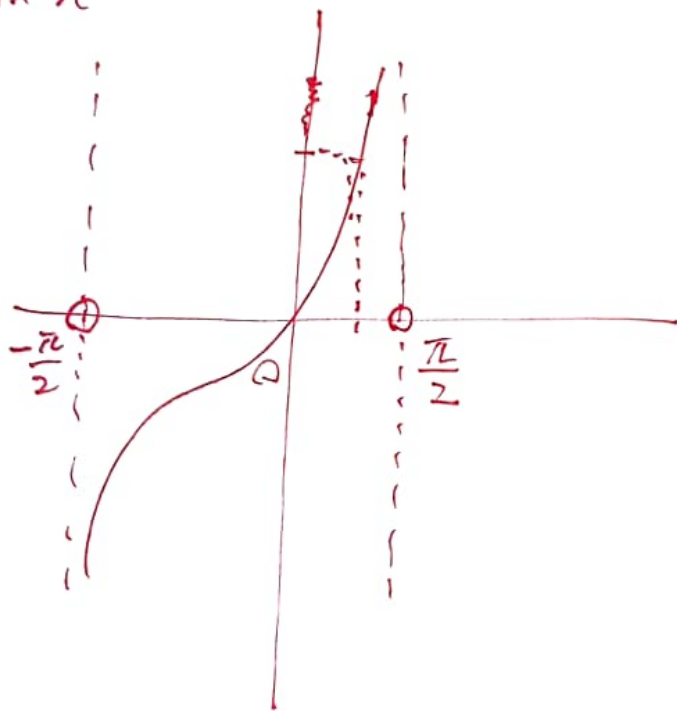
$$\begin{aligned} \lim \frac{\frac{1}{n \sqrt[n]{n}}}{\frac{1}{n}} &= \lim \frac{1}{\sqrt[n]{n}} \\ &= \frac{1}{\lim \sqrt[n]{n}} \\ &= \frac{1}{1} \end{aligned}$$

Since $\sum \frac{1}{n}$ div., $\sum \frac{1}{n \sqrt[n]{n}}$ div. by LCT.

Similar: Q49, Q50, Q52

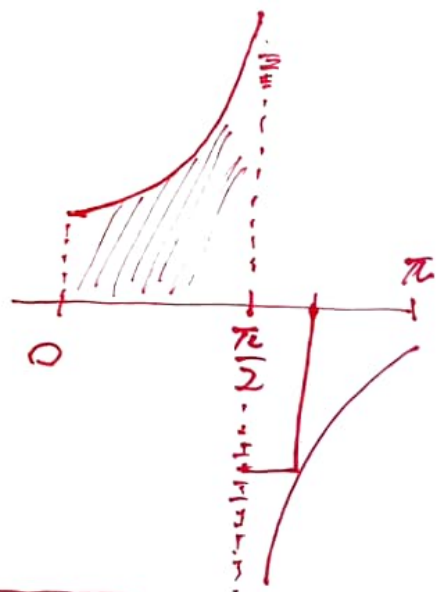
$\arctan x$

$\tan x$



$\operatorname{arcsec} x$

$$\sec x = \frac{1}{\cos x}$$



If $n \in \mathbb{N}$,

$$\Rightarrow 0 \leq \operatorname{arcsec} n < \frac{\pi}{2}$$

Valid for $n \in \mathbb{N}$
But NOT valid for $n \in \mathbb{R}$

Q 48

$$\sum_{n=1}^{\infty} \frac{\operatorname{arccsc} n}{n^{1.3}}$$

$$0 \leq \operatorname{arccsc} n < \frac{\pi}{2}$$

$$\Rightarrow 0 \leq \frac{\operatorname{arccsc} n}{n^{1.3}} < \underbrace{\frac{\pi/2}{n^{1.3}}}$$

Note that

$$0 \leq \operatorname{arccsc} n < \frac{\pi}{2} \text{ for } n \gg 1.$$

$$\frac{\pi}{2} \sum \frac{1}{n^{1.3}} \text{ conv.}$$

$$\therefore 0 \leq \frac{\operatorname{arccsc} n}{n^{1.3}} < \frac{\pi/2}{n^{1.3}}$$

Since $1.3 > 1$, so $\sum_{n=1}^{\infty} \frac{1}{n^{1.3}}$ conv. (p-series)

Since $\frac{\pi}{2} \sum_{n=1}^{\infty} \frac{1}{n^{1.3}}$ conv., $\sum_{n=1}^{\infty} \frac{\operatorname{arccsc} n}{n^{1.3}}$ conv. $\Rightarrow \frac{\pi}{2} \sum_{n=1}^{\infty} \frac{1}{n^{1.3}}$ conv.
by DCT.

Q19 $\sum \frac{\sin^2 n}{2^n}$

$\therefore$ Use DCT with $\frac{1}{2^n}$

$$0 \leq \sin^2 n \leq 1$$
$$\Rightarrow \boxed{0 \leq \frac{\sin^2 n}{2^n} \leq \frac{1}{2^n}}$$

$\downarrow$

$$\underline{\underline{\left(\frac{1}{2}\right)^n}}$$

$$\sum \left(\frac{1}{2}\right)^n \text{ conv.}$$

Geometric Series

Q28 $\sum \frac{(\ln n)^2}{n^3}$

$1 \leq \ln n$
 $1 \leq (\ln n)^2 \Rightarrow 1 \leq \frac{(\ln n)^2}{\underline{\underline{n^3}}}$

Const. $\ll \ln n \ll n^p$

$\ln n \leq n$

$\frac{(\ln n)^2}{n^3} \leq \frac{1}{n}$
 $\underline{\underline{\sum \frac{1}{n} \text{ div.}}}$

$\ln n \leq n^{\frac{1}{2}}$ for $n \geq \boxed{1}$
 $\Rightarrow \frac{(\ln n)^2}{n^3} \leq \boxed{\frac{1}{n^2}}$ $\uparrow$ Check yourself
 $\sum \frac{1}{n^2} \text{ conv.}$

$\Rightarrow$ DCT

Q30 (Hard)
 $\sum \frac{(\ln n)^2}{n^{3/2}}$